AI VIETNAM
All-in-One Course
(TA Session)

Seminar 1
Delving into Semi-Supervised Medical Segmentation

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Nguyen Thanh Huy

Outline

- Introduction to Semi-Supervised Learning
- Breakdown SSL Concepts
- Advanced SSL-SSMS Methods
- Q&A

Introduction to Semi-Supervised Learning

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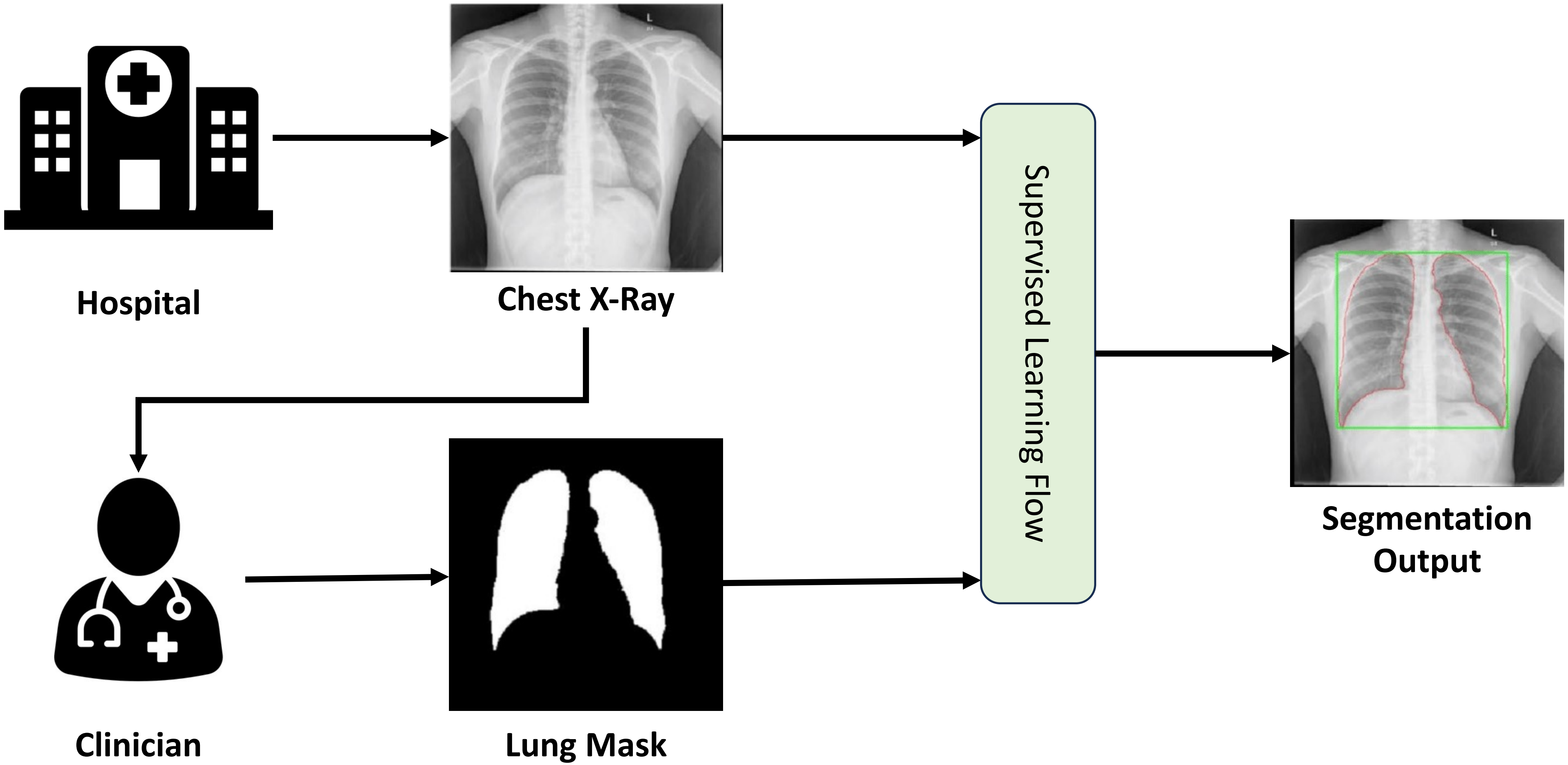
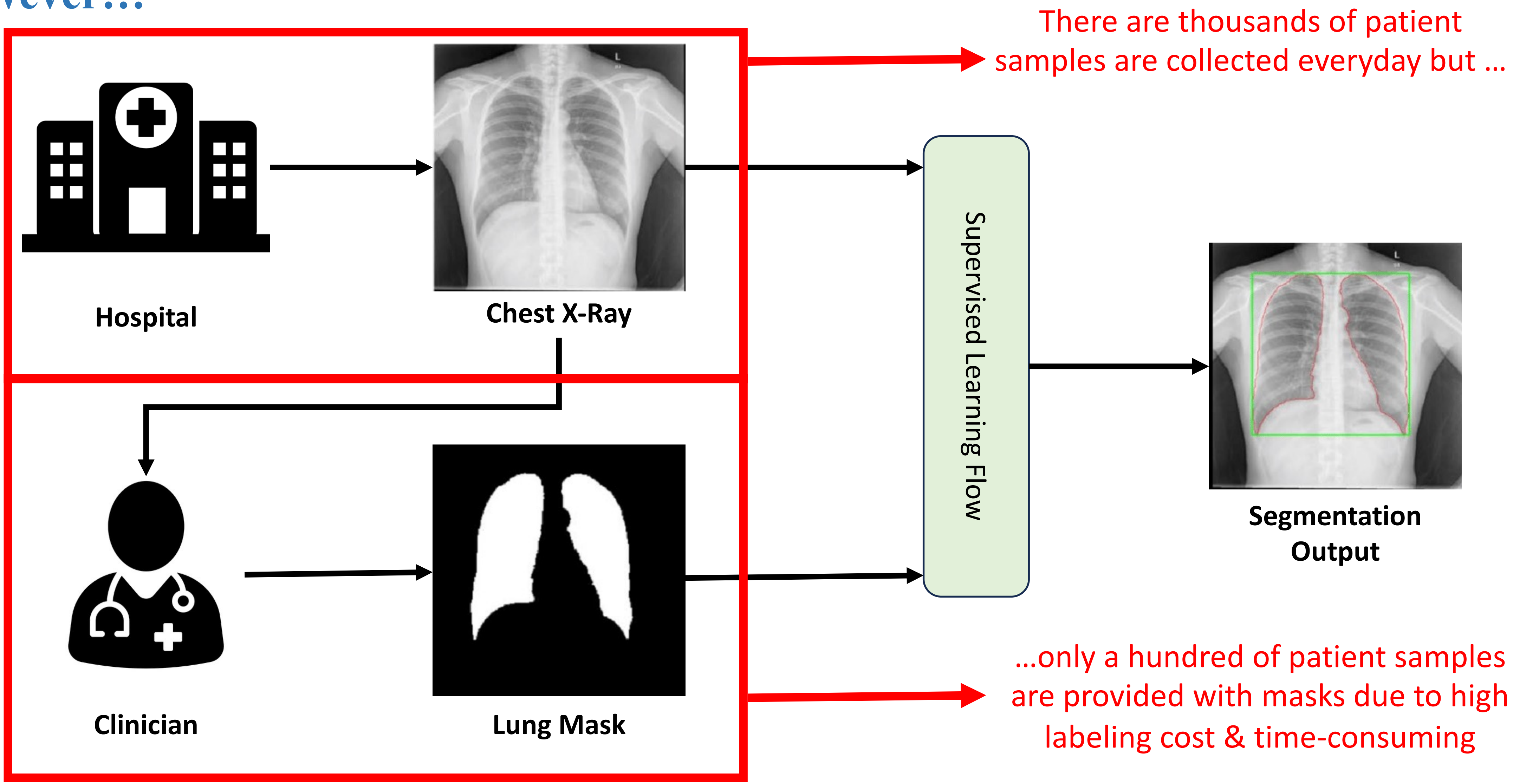
From Supervised to Semi-Supervised

❖ Supervised Learning Scenario in Real-world MedAI



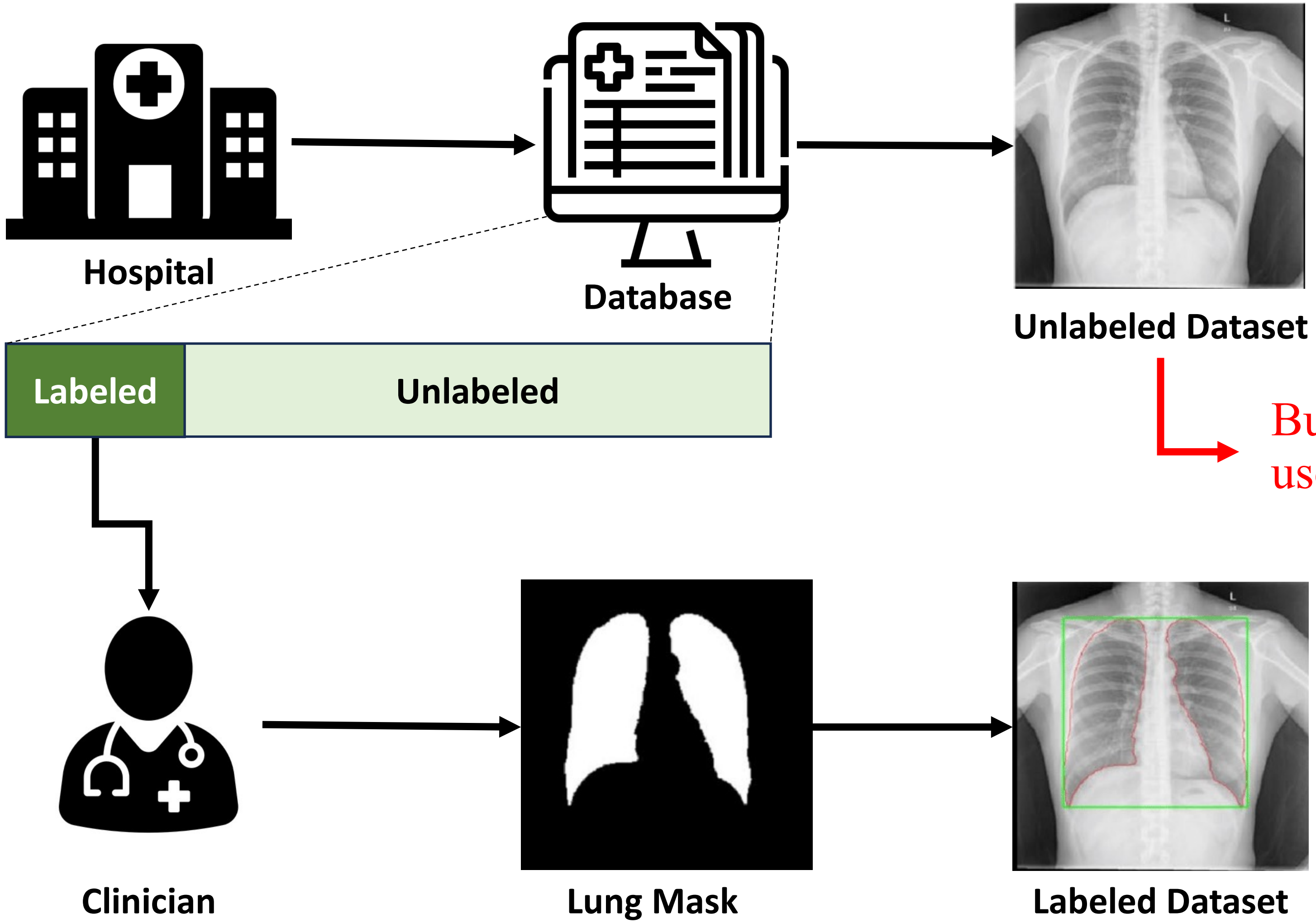
From Supervised to Semi-Supervised

However…



From Supervised to Semi-Supervised

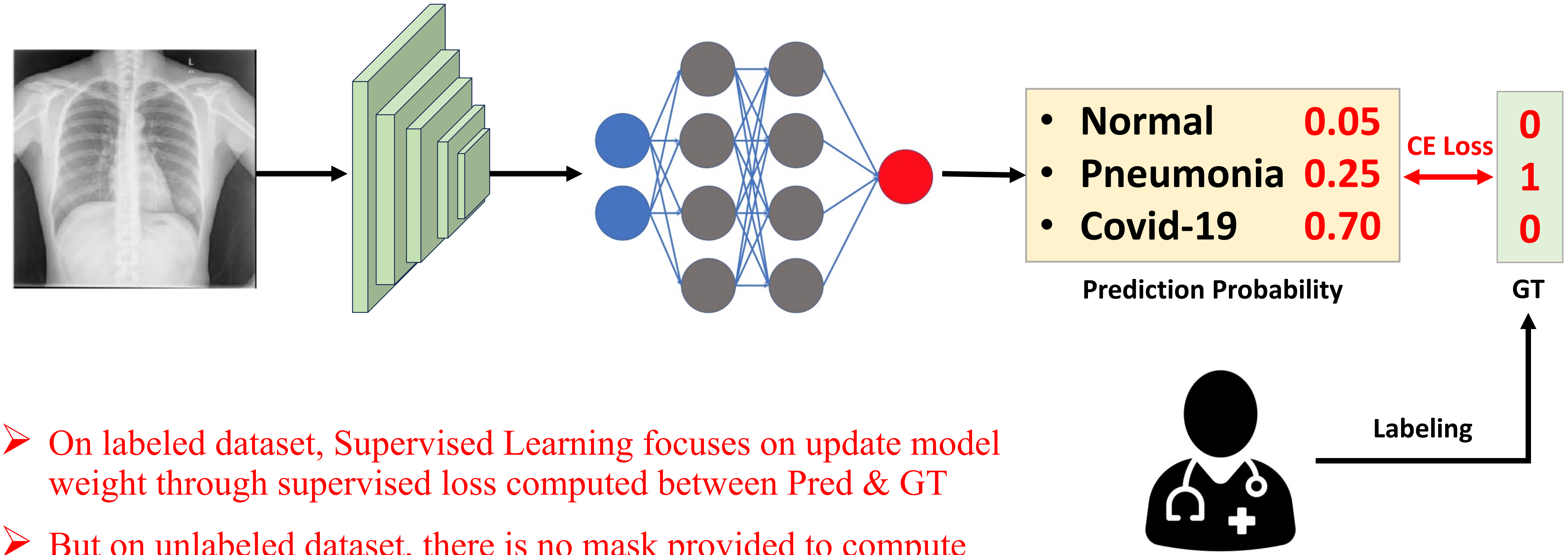
The real picture of real-world scenario would be …



But how do we effectively use the unlabeled dataset?

From Supervised to Semi-Supervised

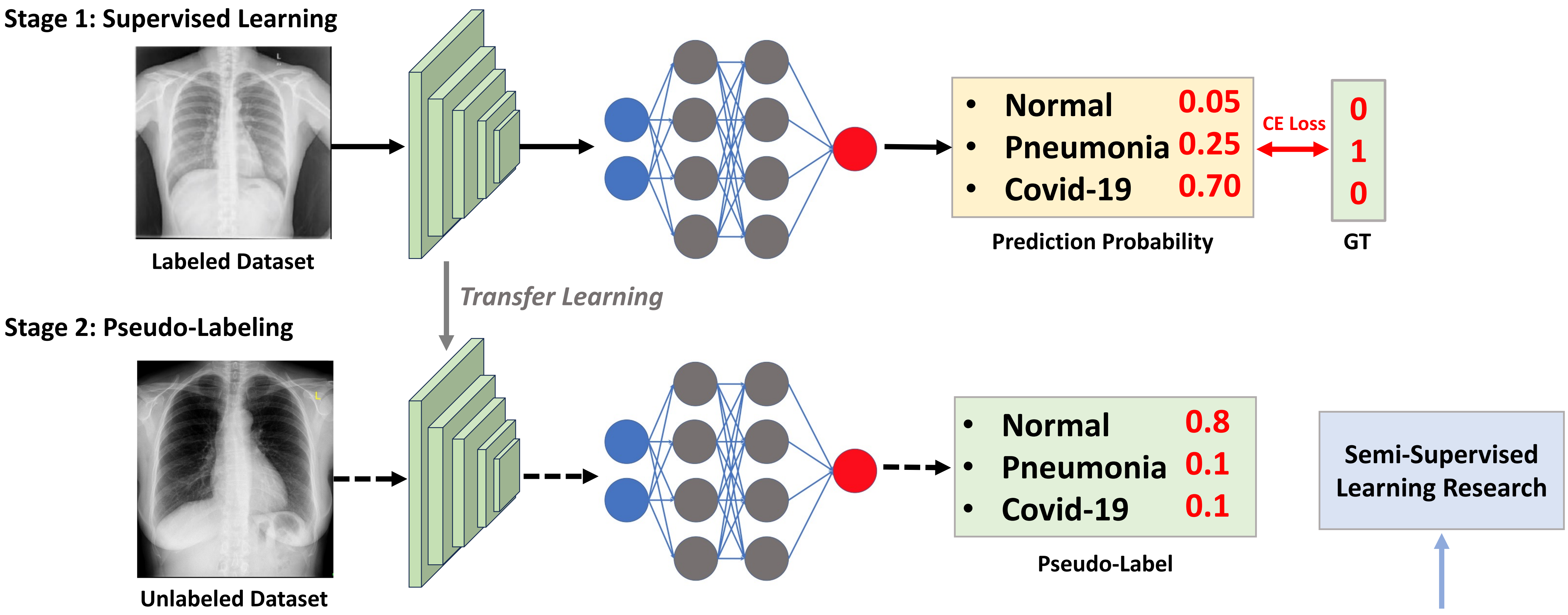
Let's look back how supervised learning update the classification model.



- On labeled dataset, Supervised Learning focuses on update model weight through supervised loss computed between Pred & GT
- But on unlabeled dataset, there is no mask provided to compute the same loss with supervised training stage.

From Supervised to Semi-Supervised

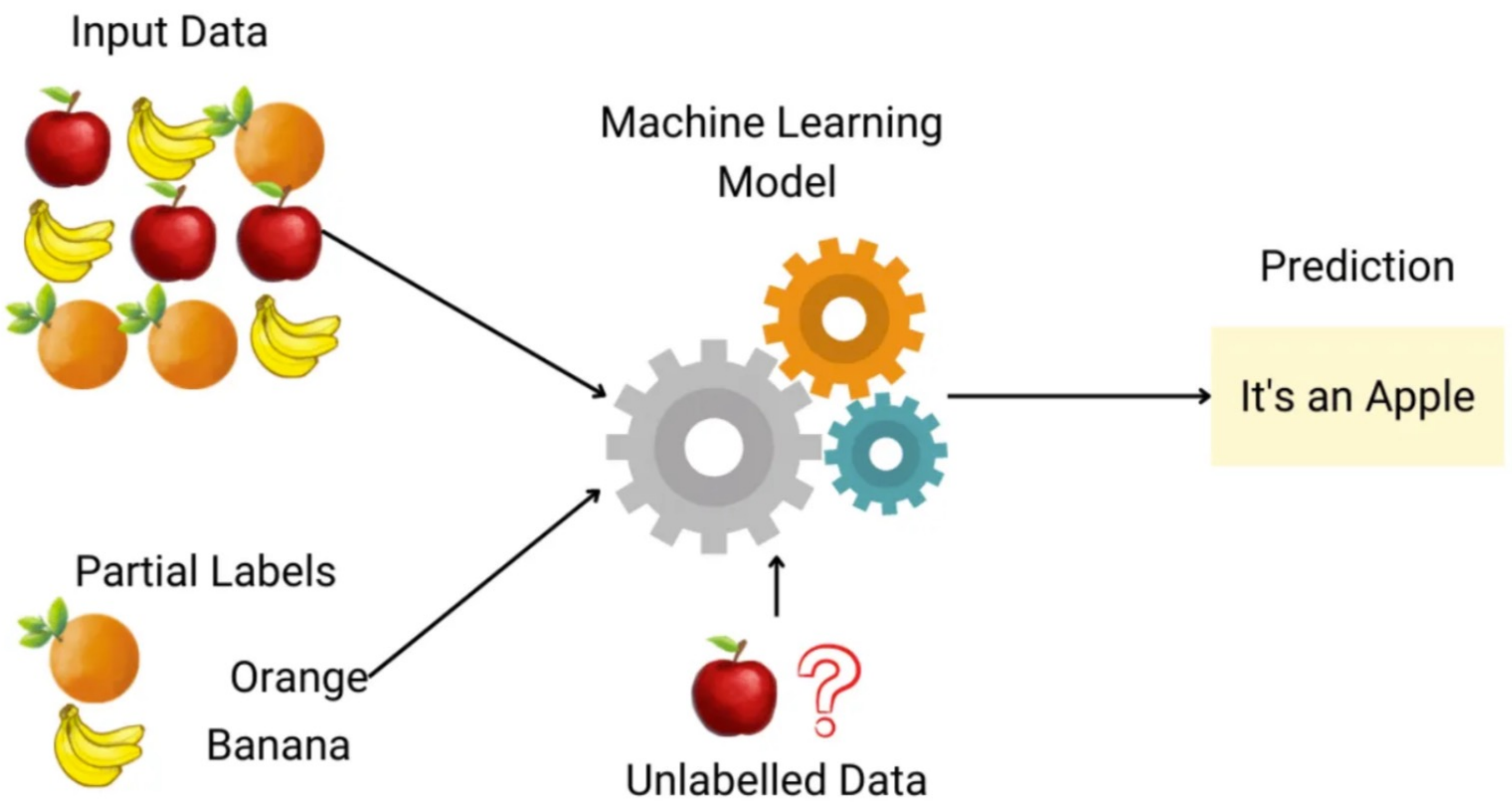
Naïve solution for unsupervised learning on unlabeled dataset – Pseudo-labeling



For unlabeled dataset, after supervised training, we use pre-trained model to predict unlabeled dataset and use that prediction as ground-truth, called **"Pseudo-Labeling"** → **Good idea to use unlabeled set But it might be not enough.**

Definition

Semi-supervised learning là một phương pháp sử dụng kết hợp cả dữ liệu có nhãn và dữ liệu không nhãn trong quá trình huấn luyện mô hình học máy.



Whole picture of SSL setting

When **facing a limited amount of labeled data for supervised learning tasks**, four approaches are commonly discussed:

1. Pre-training + fine-tuning
2. Semi-supervised learning.
3. Active learning
4. Pre-training + dataset auto-generation

Hypotheses of SSL

Several hypotheses have been discussed in literature to support certain design decisions in semi-supervised learning methods.

H1-Smoothness Assumptions: If two data samples are close in a high-density region of the feature space, their labels should be the same or very similar.

H2-Cluster Assumptions: The feature space has both dense regions and sparse regions. Densely grouped data points naturally form a cluster. Samples in the same cluster are expected to have the same label. This is a small extension of H1.

H3-Low-density Separation Assumptions: The decision boundary between classes tends to be located in the sparse, low density regions, because otherwise the decision boundary would cut a high-density cluster into two classes, corresponding to two clusters, which invalidates H1 and H2.

H4-Manifold Assumptions: The high-dimensional data tends to locate on a low-dimensional manifold. Even though real-world data might be observed in very high dimensions (e.g. such as images of real-world objects/scenes), they actually can be captured by a lower dimensional manifold where certain attributes are captured and similar points are grouped closely (e.g. images of real-world objects/scenes are not drawn from a uniform distribution over all pixel combinations). This enables us to learn a more efficient representation for us to discover and measure similarity between unlabeled data points. This is also the foundation for representation learning.

Breakdown SSL Concepts

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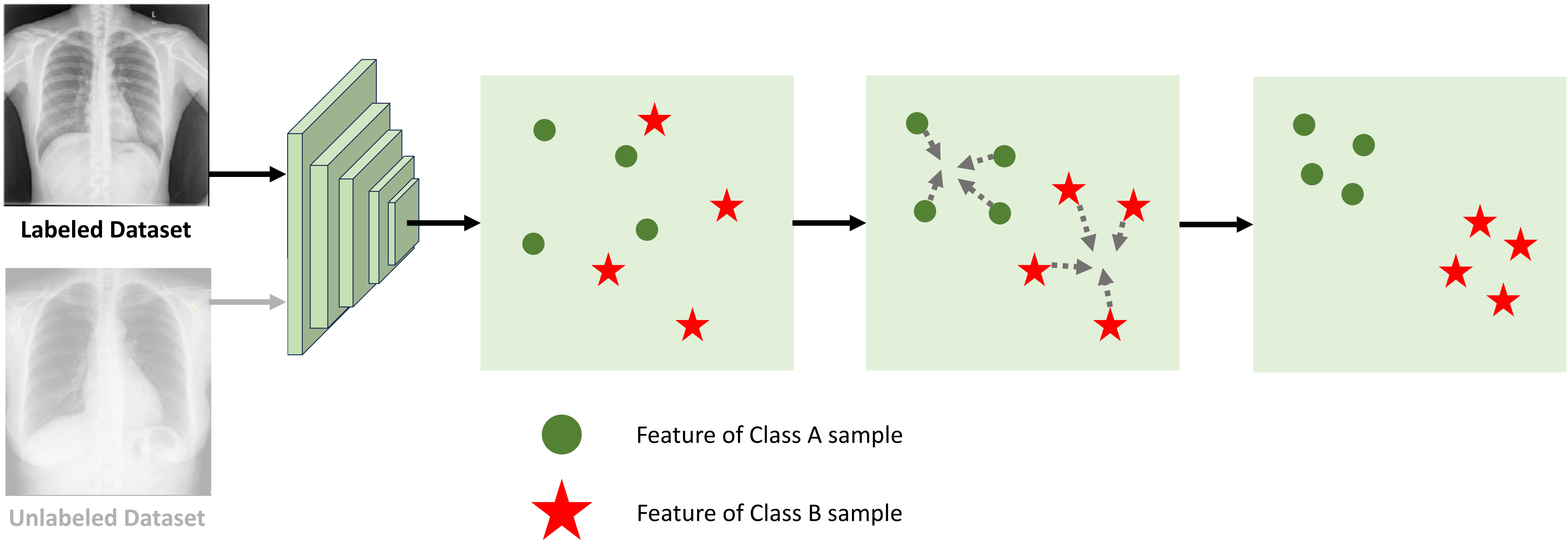
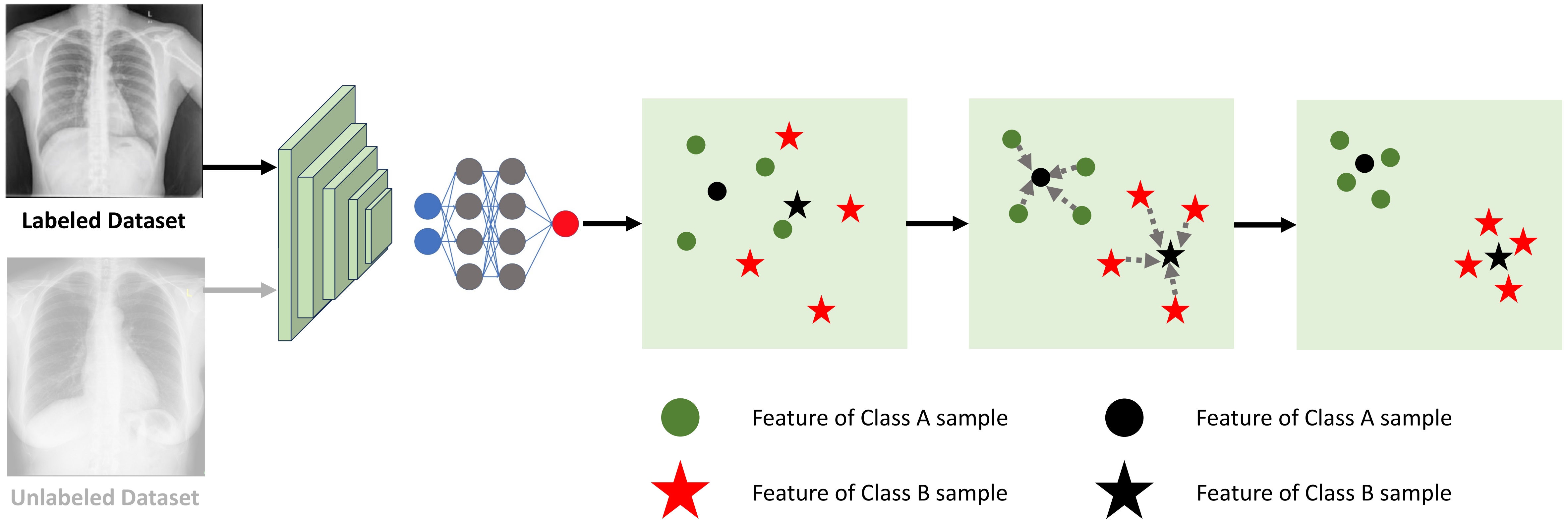
Looking deeper into SSL setting

Supervised Learning (CNN – Feature Extraction Representation)



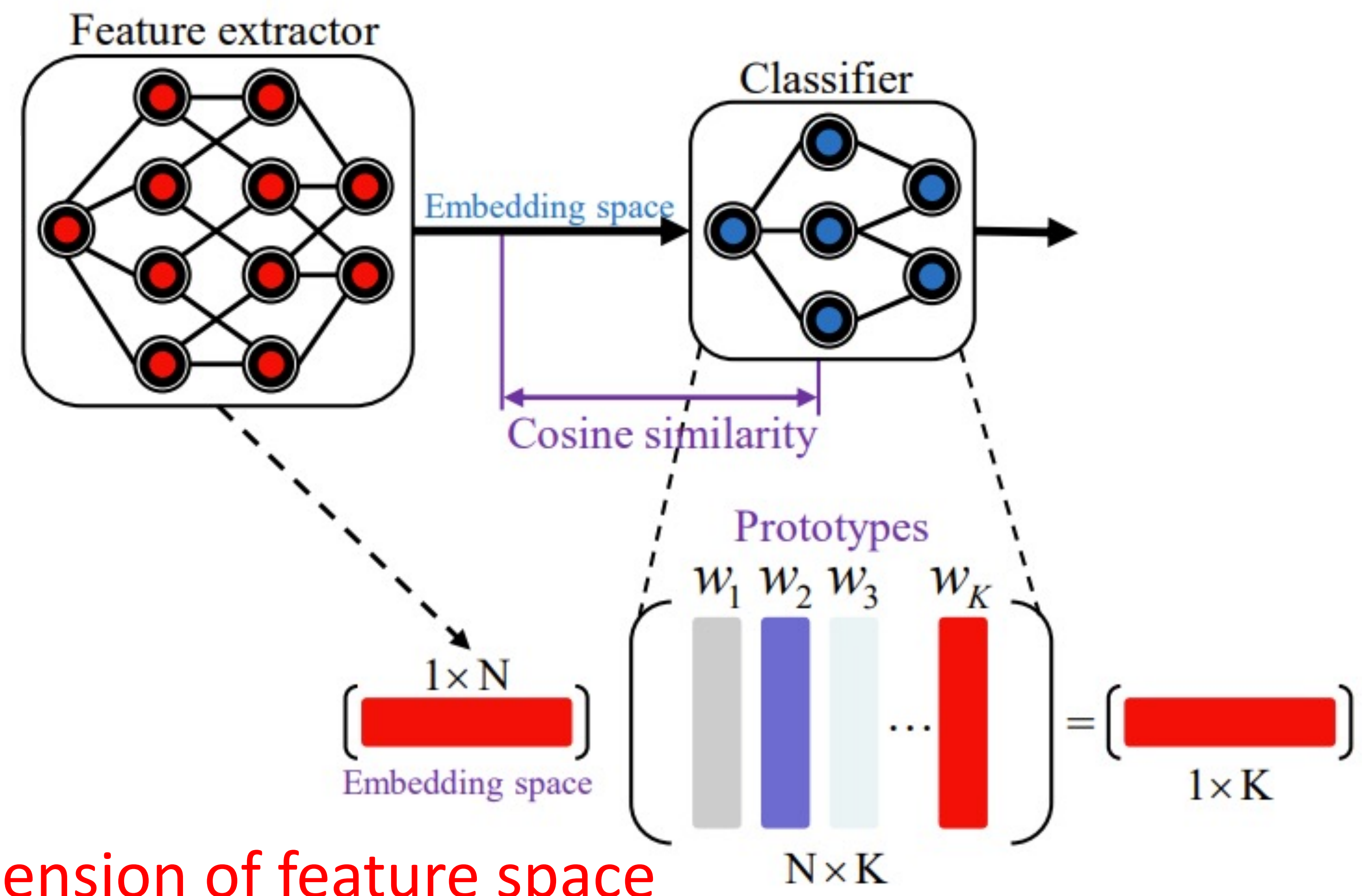
Looking deeper into SSL setting

Supervised Learning (MLP – Classifier Representation)



Looking deeper into SSL setting

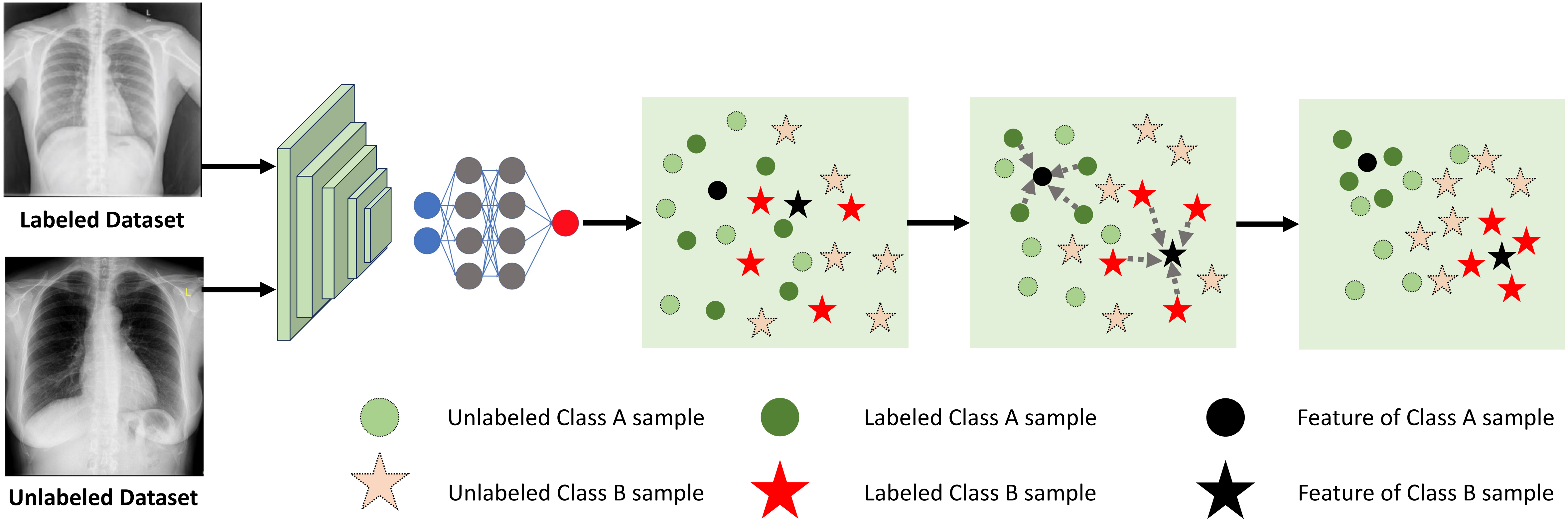
Concept behind feature representation



- ➢ N is the dimension of feature space
- ➢ K is the number of classes

Looking deeper into SSL setting

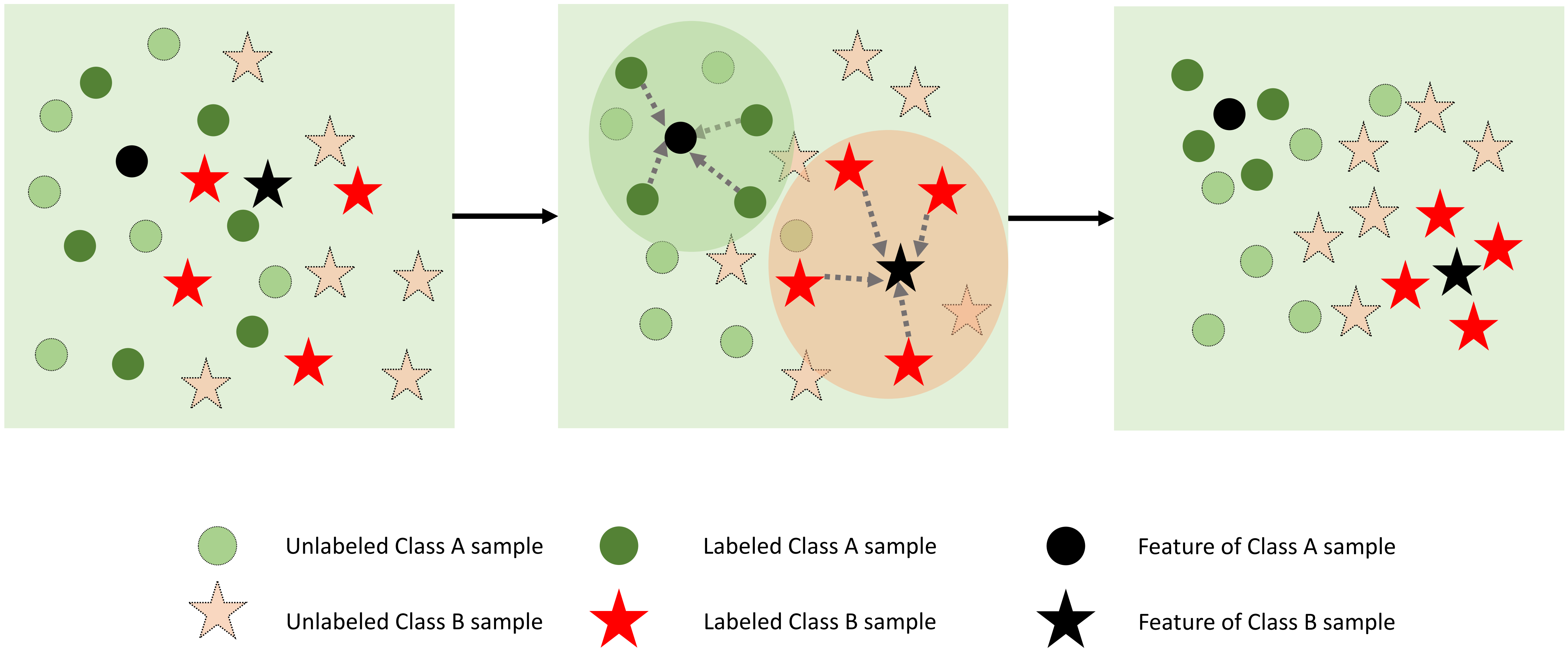
Semi-Supervised Learning (Feature representation)



➢ **What is the problem?**

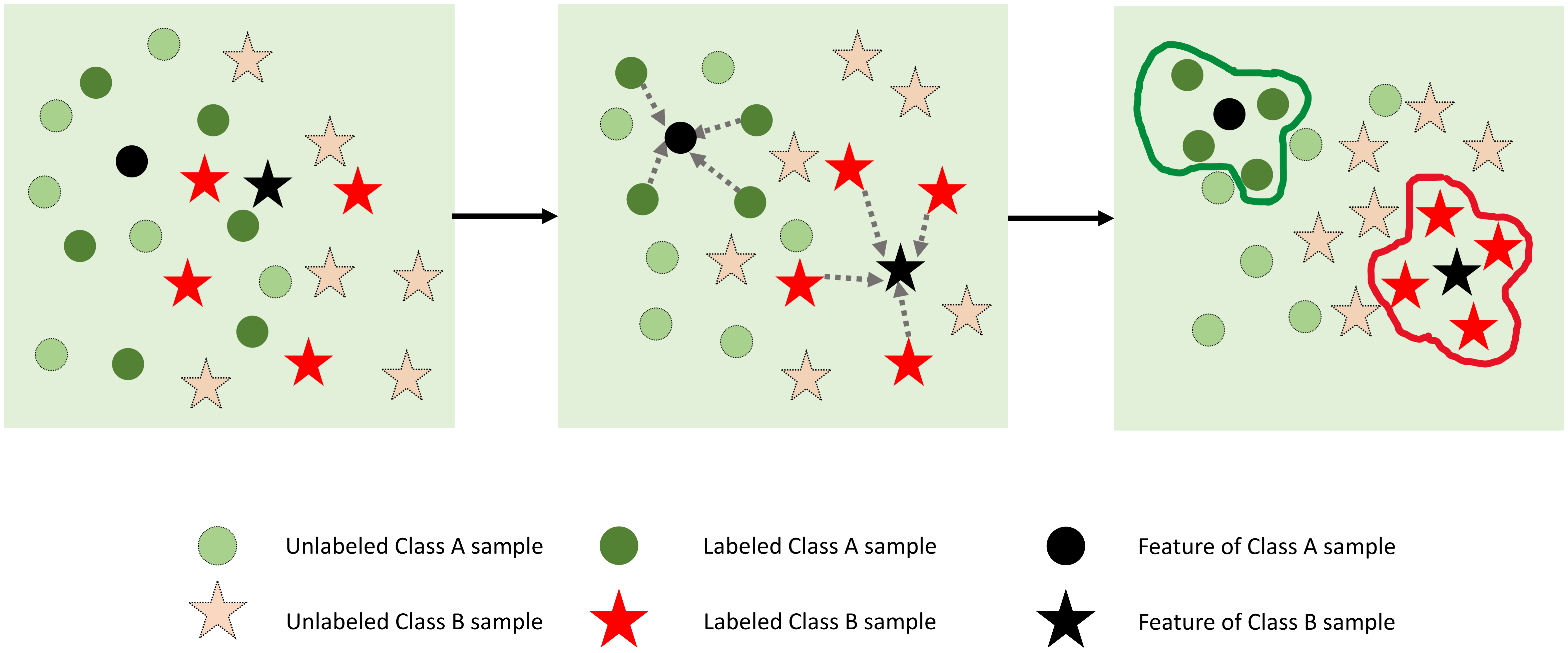
Looking deeper into SSL setting

Problem 1: Feature Extraction tends to be bias due to lack of information



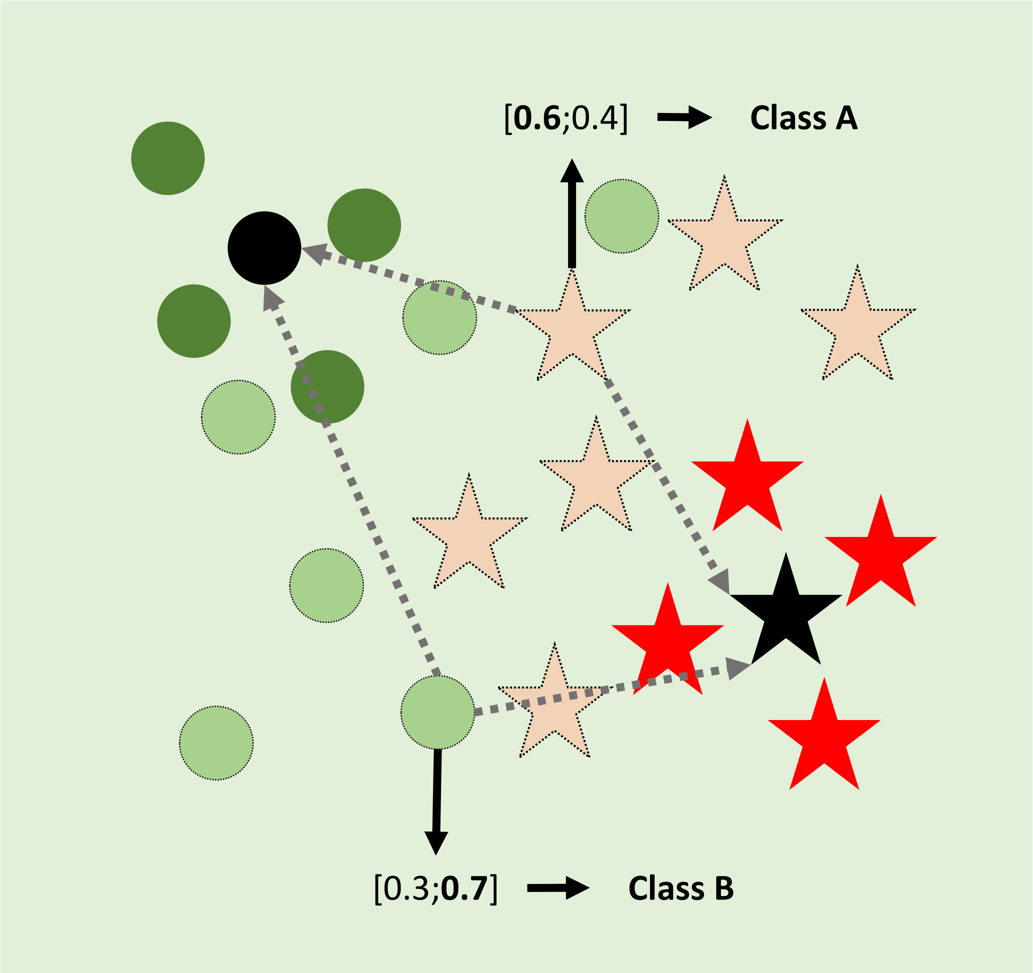
Looking deeper into SSL setting

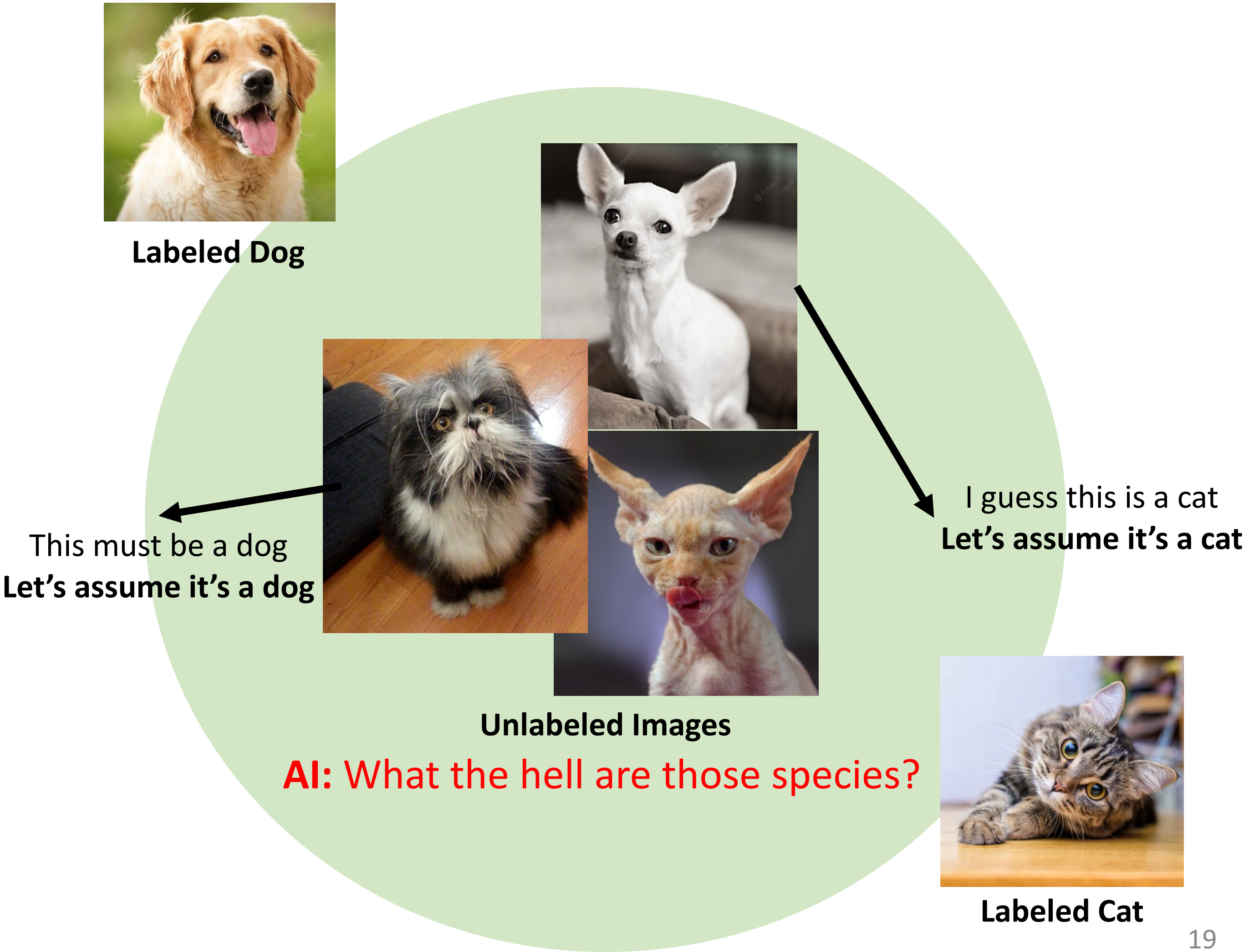
Problem 2: Prototype does not represent based on all data points



Looking deeper into SSL setting

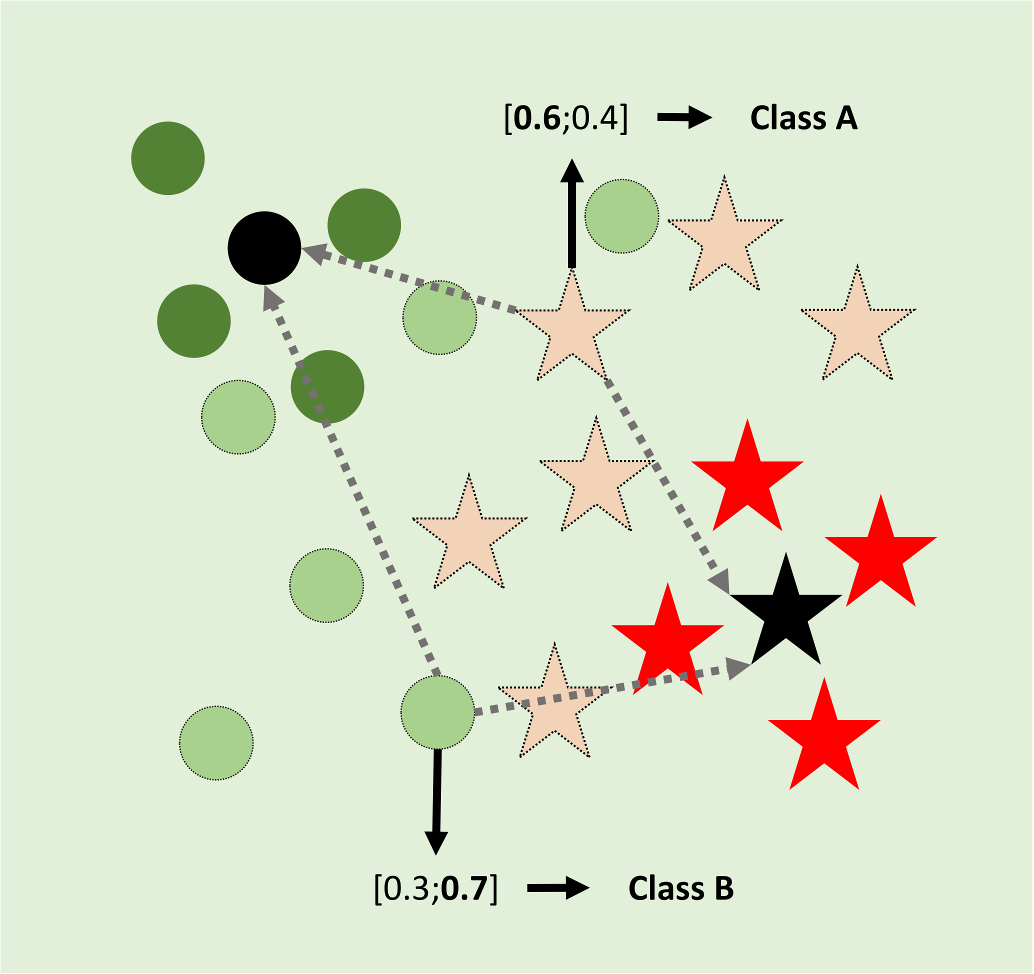
Consequence 1: Hard unlabeled samples become the noisy pseudo-labeled samples

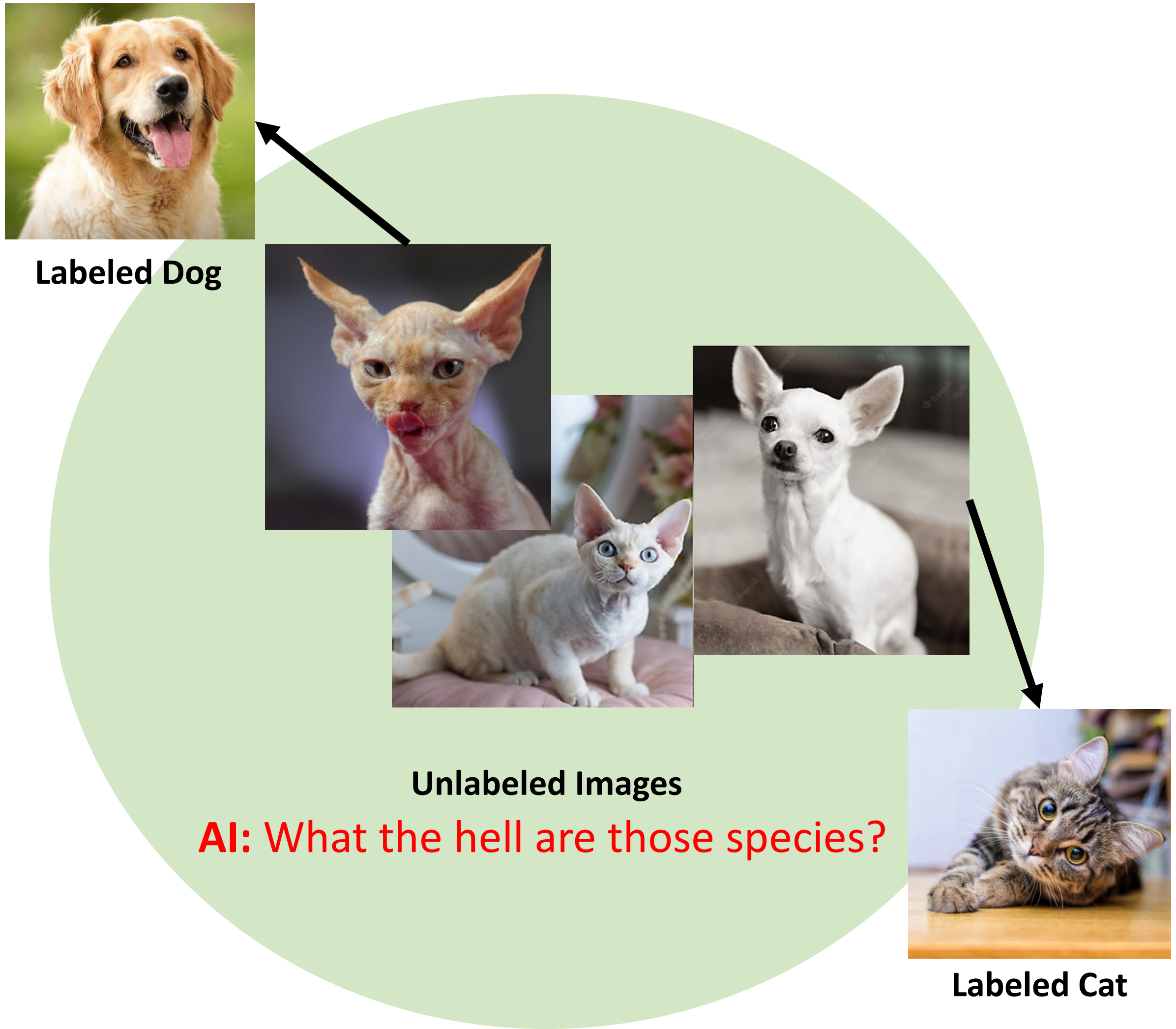




Looking deeper into SSL setting

Consequence 2: Test images are failed due to train set is too small to represent classes information





Advanced SSL-SSMS Methods

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Consistency Regularization

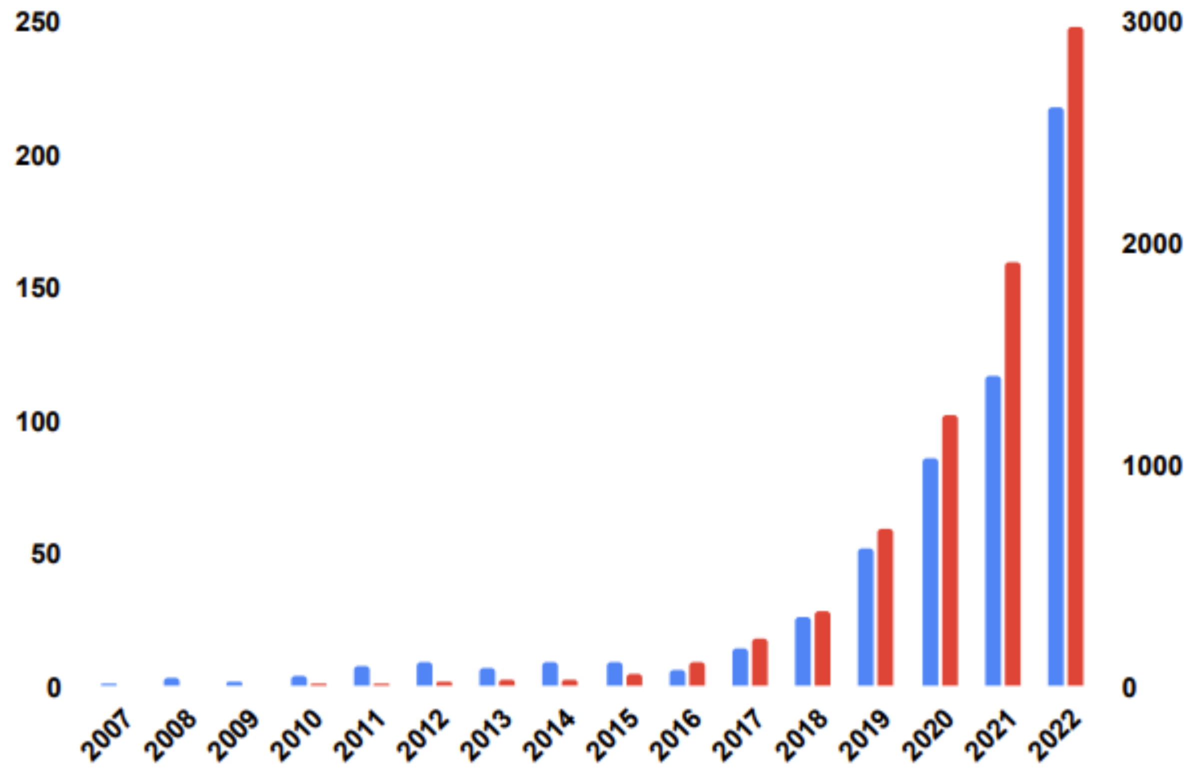
FixMatch

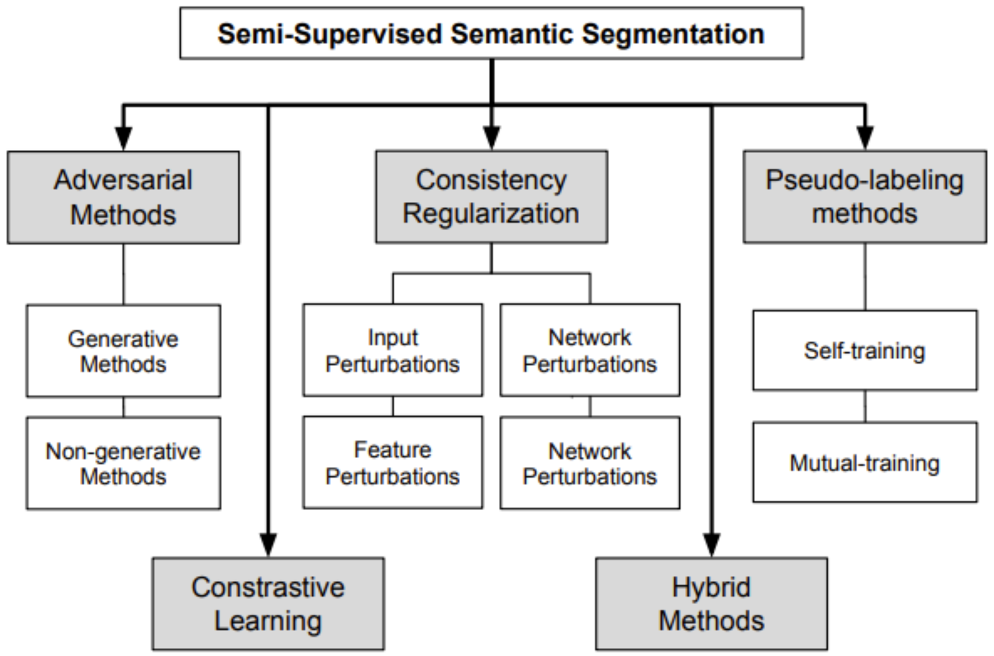
Limitation of FixMatch

FixMatch -> Dash/FlexMatch/FreeMatch

Semi-Supervised Semantic Segmentation

Semantic segmentation is one of the most challenging tasks in computer vision. However, in many applications, a frequent obstacle is the lack of labeled images, due to the high cost of pixel-level labeling. In this scenario, it makes sense to approach the problem from a semi-supervised point of view, where both labeled and unlabeled images are exploited.





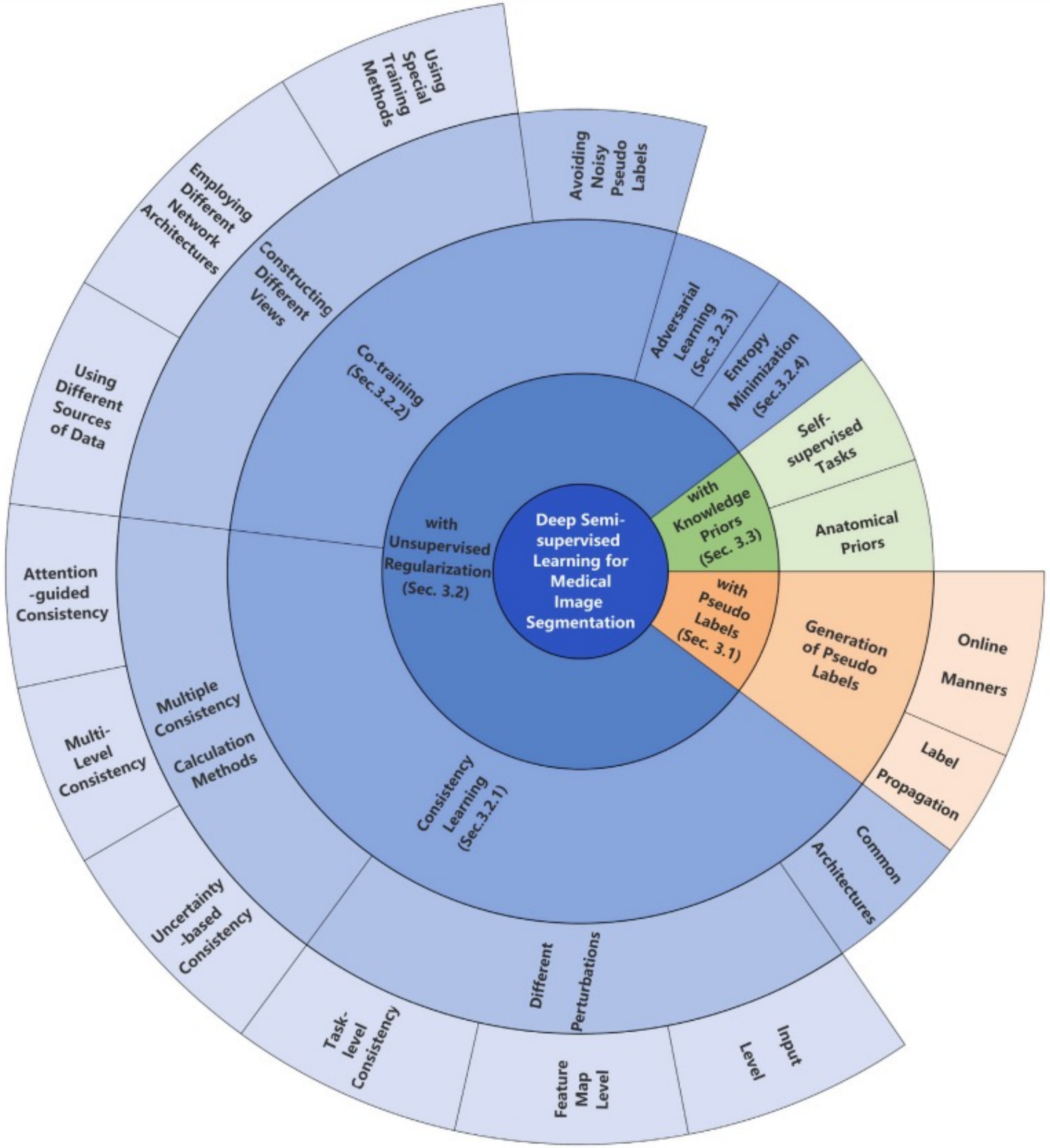
Feature Pertubation

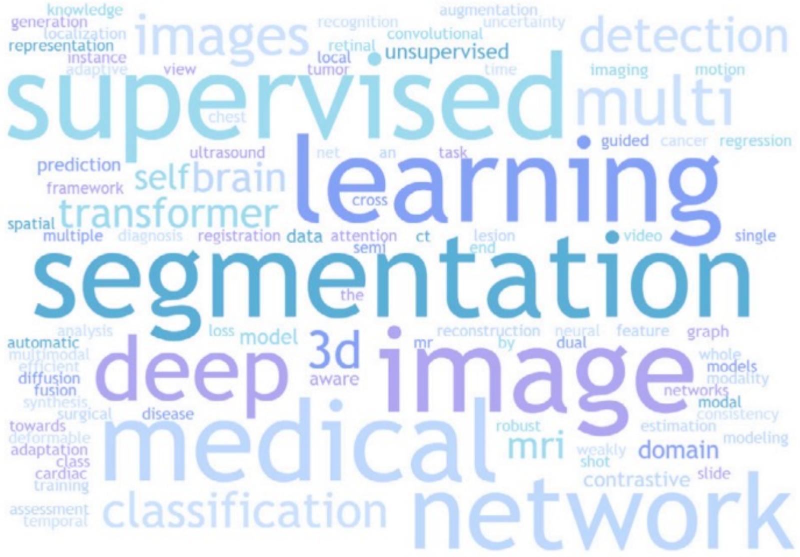
CCT

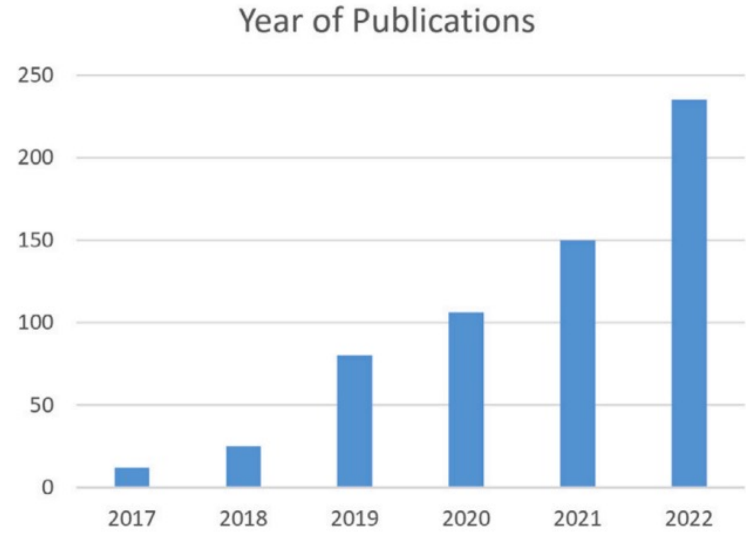
Co-Training

FixMatch -> Dash/FlexMatch/FreeMatch

All Branches of Semi-Supervised Medical Segmentation







Q&A

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Questions & Answers

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Thank you!